MILIND GUNJAL

PhD in Mathematics | Aspiring Quantitative Researcher

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PROFESSIONAL SUMMARY

PhD in Pure Mathematics with 6+ years of experience in advanced mathematical research, teaching, and data analysis. Strong foundation in probability theory, statistical modeling, and programming (Python, R, MATLAB). Proven ability to work with complex abstract systems and translate them into computational models. Seeking to leverage analytical and programming skills in quantitative finance roles.

EDUCATION

FLORIDA STATE UNIVERSITY — **TALLAHASSEE, FLORIDA**

MS in Mathematics (CGPA: 3.91/4) | PhD in Mathematics — 2019-2025

- **PhD Thesis:** Studied Monoidal 2-Categories, and Algebraic K-theory (Reviewed 110+ research papers and books to work on 3 projects in the field of homotopy theory and category theory. Axiomatized algebraic models for the K-theory of the generalized notion of algebraic gadgets like rings, spaces, varieties, called Waldhausen categories)
- **Relevant Courses:** Measure and Probability Theory, Logic, Type Theory, and the Mechanization of Mathematics

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH — **KOLKATA, INDIA**

MS in Mathematics and Statistics (BS-MS Dual Degree Course; KVPY Fellow, Valedictorian) — 2014-2019

- **MS Thesis:** Studied Line Bundles from Algebraic viewpoint (Set up foundations for comparing two areas of mathematics—algebraic K-theory and topological K-theory using Serre-Swan theorem)
- **Projects**
  - Modules over the ring of continuous functions: Explored algebraic structures and their topological implications
  - Representation Theory of GL(2,F_q): Investigated finite group representations using field theory and character tables

PROJECTS IN QUANT FINANCE

- **SABR Stochastic Volatility Model: Calibration, Greeks, and Copula-Based Multi-Asset Pricing:** Calibrated SABR (α, β, ρ, ν) to live SPY data; computed full Greeks (Delta–Volga) via finite differencing with a delta hedging back test; priced worst-of/best-of exotic options on a two-asset basket using Gaussian, Student's t, and Clayton copulas via Monte Carlo.
- **Trinomial Tree Option Pricing:** Extended binomial backward induction to a trinomial framework, enabling American option pricing with early exercise features; computed all Greeks (Delta, Gamma, Theta via tree nodes; Vega, Rho via bumping) and validated convergence to Black-Scholes-Merton values.
- **Modelling USDEUR FX Swaptions:** Compared Black '76 vs. Merton Jump-Diffusion for a cross-currency swaption (yield curve bootstrapping + Vanna-Volga interpolation); quantified an 8.06% jump risk premium.
- **Quantitative Momentum Trading & VaR Optimization:** Built a systematic risk engine (Historical, Parametric, and Monte Carlo VaR) with Kupiec LR and Traffic Light back testing on NSE momentum strategies; applied Markowitz MVO to transition from equal-weighting to risk-optimized allocation, benchmarked via Sharpe, Alpha, Beta and Max Drawdown.

CERTIFICATIONS AND OTHER COURSES

Quant Finance Boot Camp certificate program by the Erdős Institute: Implemented GBM for correlated multi-asset portfolios (Cholesky decomposition) with Monte Carlo VaR; derived and applied Black–Scholes to recover implied volatility from live TSLA option chains; calibrated the Heston stochastic volatility model (Fourier inversion) to market prices; back tested delta hedging on a real two-year price history.

TECHNICAL SKILLS

- **Mathematical & Analytical:** Mathematical modeling, Statistical inference, Topological data analysis
- **Programming & Tools:** Python, R, MATLAB, C++, Git, LaTeX, Advanced Excel
- **Quantitative Finance & Statistical Methods:**
  - Time Series: ARIMA, ARCH, GARCH, GARCHX
  - Derivatives & Risk: Itô's Calculus, Black-Scholes model, Heston model, SABR model, Trinomial trees, Value at Risk
  - Probability & Statistics: Martingale Systems, Copula Methods, Regression, Stochastic Process, Monte Carlo Simulation

PROFESSIONAL EXPERIENCE

TALLAHASSEE STATE COLLEGE — **TALLAHASSEE, FLORIDA**

Adjunct Faculty — 2025-Present

Taught advanced undergraduate math courses (Multivariate Calculus, Linear Algebra) for 110 students every semester; designed curriculum, and analyzed performance data to improve instruction

MERCOR, HANDSHAKE — **REMOTE**

Mathematics Expert — 2025-Present

Expertly trained and refined advanced LLMs to enhance their capability in solving and explaining mathematical problems. Authored and meticulously reviewed hundreds of complex math prompts, solutions, and explanatory rationales across diverse fields including abstract algebra, probability theory

FLORIDA STATE UNIVERSITY — **TALLAHASSEE, FLORIDA**

Administrative Teaching Assistant — 2023-2025

Taught advanced undergraduate courses (Multivariate Calculus, Linear Algebra) for 30–100 students; designed curriculum, managed TAs, and analyzed performance data to improve instruction, led a team of 20 volunteers for the AMS sectional meeting with 500+ attendees, optimized workspace assignments for 100+ TAs under logistical constraints, improving operational efficiency

Instructor of Record/Lab Instructor — 2019-2023

Designed and taught multivariate calculus and linear algebra courses, focusing on real-world applications and computational methods used in quantitative modeling. Analyzed student performance data using Excel to identify trends and optimize instructional methods

SELECTIVE PUBLICATIONS

- Symmetric Monoidal Bicategories and Biextensions (2024)(Reviewed 30+ research papers and books to analyze symmetric monoidal bicategories using algebraic models called biextensions)
- On a 2-determinant functor of Waldhausen Categories (In preparation, expected in 2026)
- Multiplicative comparison of Waldhausen and Segal K-theory using Chain Complexes (In preparation, expected in 2026)

SUMMER INTERNSHIP/ TALKS

- Talks: 9 different talks presenting advanced mathematical research throughout the US — 2021-2024
- Internship: Study of Simplicial and Singular Homology at CHENNAI MATHEMATICAL INSTITUTE — 2018
- Project: Evolutionary Game Theory at INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH, KOLKATA — 2017

AWARDS

- Dwight B. Goodner Mathematics Fellowship (Florida State University, 2024)
- Distinguished Teaching Assistant award (Florida State University, 2023)
- Director's Gold Medal (Awarded to the Department Topper at the end of the 5-year BS-MS program) (IISER-Kolkata, 2019)
- KVPY Fellow (Department of Science and Technology, Ministry of Human Resource Development, India, 2014-2019)

EXTRA-CURRICULAR

- Students' Affairs Council General Secretary (Sports) (2017-2018, IISER, Kolkata)
- Treasurer, Inquivesta 2016 (College fest, IISER, Kolkata)
- Captained Inter IISER Sports Meet for Gold (2018, Kabaddi), Gold (2016) & Silver (2017) in Volleyball